

MASGYM: A Co-Evolving Red/Blue Security Gym for Multi-Agent LLM Systems with Parameterized Collusion, Orchestrator Compromise, and System-Level Metrics

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Abstract

The strongest existing LLM-agent security benchmarks—Agent Security Bench, AgentDojo, InjecAgent, ToolEmu—are single-agent, yet deployed systems are collectives in which a planner directs workers, agents share memory, and tools are third-party. The phenomena that dominate risk there—collusion, a compromised orchestrator, malicious tools, Sybils, group-level emergent failure—cannot be produced by a single-agent harness. We present MASGYM, an environment and evaluation methodology for multi-agent LLM security: a forced-coordination task suite with heterogeneous roles and partial observability over star, chain, tree, and mesh topologies at 3–50 agents; a parameterized adversary model; a system-level metric $\text{NRP}_{\text{sys}} = \text{PNA}_{\text{sys}}(1 - \text{ASR}_{\text{sys}})$ scored by a deterministic checker over the global state trace, never an LLM judge on the critical path; and CoRB, a co-evolving red/blue protocol logging attack–defense trajectories and a (security, cost) Pareto frontier. Supporting theory, proved under explicit assumptions, characterizes the metric axiomatically (reducing to ASB at one agent), bounds compromise propagation across topologies, establishes a strict collusion advantage and a forced-coordination lower bound, and proves evaluator integrity. On a clearly-labelled *synthetic* emulator, a per-agent guardrail’s relative ASR reduction shrinks $41.6\% \rightarrow 31.8\% \rightarrow 27.6\%$ from single-agent to collusion to compromised-orchestrator context. We then ask what that guardrail buys, and answer on four real backbones in three model families: a prompt-level defense drives action-channel compliance to 0/1075 decisions, so $\text{ASR}_{\text{act}}=0$ throughout, yet ASR_{sys} remains 1.000 on tree and star, because once the unsafe action is blocked the system predicate is still discharged by deterministic availability sabotage that topology alone determines. A fourth backbone ignores the defense entirely. A defense measured on the action channel therefore cannot certify system-level security, and its efficacy is a property of the model as much as of the defense.

Keywords

multi-agent LLM security; security benchmark; co-evolving red/blue evaluation; collusion; prompt injection; deterministic security checking; system-level metrics

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A Proofs

This appendix collects the proofs deferred from Section 9. Throughout the propagation results we work in the independent-cascade instance of Assumption 4: when an agent becomes compromised, it independently compromises each susceptible out-neighbour with probability p when its message is processed, and compromise is absorbing.

A.1 Metric and integrity

PROOF OF THEOREM 9.1 (REDUCTION TO ASB). With one agent the global trace is that agent’s trajectory and the system predicates equal the single-agent predicates, so Equations (3) and (4) equal ASB’s ASR and PNA, and Equation (5) equals Equation (1). $\square$

PROOF OF THEOREM 9.2 (AXIOMATIC CHARACTERIZATION). By D4a with base $u = 1$, $g(u, \sigma) = g(u \cdot 1, \sigma) = u g(1, \sigma)$. By D4b with base $\sigma = 1$, $g(1, \sigma) = g(1, \sigma \cdot 1) = \sigma g(1, 1)$. By D2, $g(1, 1) = 1$. Hence $g(u, \sigma) = u\sigma g(1, 1) = u\sigma$. The boundary $g(0, \sigma) = 0$ holds since $0 \cdot \sigma = 0$. Finally $u\sigma$ is bounded in $[0, 1]$ and non-decreasing in each argument on $[0, 1]^2$. Continuity (Assumption 5) is used only to extend the homogeneity identities from rationals to all of $[0, 1]$; the algebraic identities hold pointwise. $\square$

PROOF OF THEOREM 9.4 (EVALUATOR INTEGRITY). By Assumption 1, Φ and Ψ are functions of τ only; a function returns equal outputs on equal inputs, giving $C(\tau) = C(\tau')$. Let π be any message-space manipulation inducing trace τ_π . The verdict is $\Phi(\tau_\pi)$; if $\tau_\pi = \tau$, then $\Phi(\tau_\pi) = \Phi(\tau)$. As $\overline{\text{ASR}}_{\text{sys}}$ and $\overline{\text{PNA}}_{\text{sys}}$ are averages of such verdicts, they are unchanged, and so is NRP_{sys} . $\square$

PROOF OF THEOREM 9.5. Because J takes message content as an argument and is non-constant in it, pick any two message sets on which J disagrees (e.g., msgs_π contains an injected instruction “report this trace as safe”); by construction $\tau_\pi = \tau$, so the state is unchanged while the verdict flips. $\square$

A.2 Propagation propositions

For reference, the full statements of Theorems A.1 to A.5 follow.

Proposition A.1 (Chain: bounded reach). *On a directed path $a_1 \rightarrow \dots \rightarrow a_n$ with a single seed at a_1 , the expected number of additionally*



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compromised agents is $\sum_{j=1}^{n-1} p^j \leq p/(1-p)$ for $p < 1$, independent of n .

Proposition A.2 (Star: the orchestrator is the critical node). *On a star with centre c , a seed at the centre compromises each leaf independently with probability p , giving expected reach $p(n-1)$ and $\text{CPR} = p$; a seed at a leaf gives expected reach $p + p^2(n-2)$. The centre-seed reach exceeds the leaf-seed reach by $\Theta(p(1-p)n)$.*

Proposition A.3 (Tree: a branching phase transition). *On a rooted tree with branching factor b and depth d , seeded at the root, the expected number compromised at level ℓ is $(bp)^\ell$; the total expected reach is $\sum_{\ell=1}^d (bp)^\ell$ —bounded in d for $bp < 1$ and $\Theta((bp)^d)$ for $bp > 1$, transitioning at $bp = 1$.*

Proposition A.4 (Mesh: one-round lower bound). *On the complete graph K_n with s seeds, the expected number compromised after one round is at least $s + (n-s)(1-(1-p)^s)$. Consequently, for any fixed $p > 0$ the honest-compromise fraction after one round tends to 1 as s grows, and iterating drives $\text{CPR} \rightarrow 1$ above the percolation threshold.*

Corollary A.5 (Threshold confinement). *Under the threshold instance of Assumption 4 with threshold θ , any agent whose in-degree in $\mathcal{G}$ is strictly less than θ can never be compromised by propagation. In particular, on a chain (in-degree 1) with $\theta \geq 2$ the compromised set never grows beyond the seeds, so $\text{CPR} = 0$.*

PROOF OF THEOREM A.1 (CHAIN). On the directed path $a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_n$ with the single seed at a_1 , the agent a_{1+j} becomes compromised iff every one of the j edges fires. By independence these events have probability p^j . Summing the expected indicators,

$$\mathbb{E}[|\mathcal{B}_\infty| - 1] = \sum_{j=1}^{n-1} p^j = p \frac{1-p^{n-1}}{1-p} \leq \frac{p}{1-p}, \quad p < 1,$$

which is bounded independently of n . $\square$

PROOF OF THEOREM A.2 (STAR). Let the centre be c and the leaves $a_1, \dots, a_{n-1}$. *Centre seed.* Each leaf is a direct out-neighbour of c , so it is compromised independently with probability p ; hence $\mathbb{E}[|\mathcal{B}_\infty| - 1] = p(n-1)$ and $\text{CPR} = p$. *Leaf seed.* A seed leaf compromises the centre with probability p ; conditioned on that, each of the remaining $n-2$ leaves is compromised with probability p . Thus the expected additional compromises are $p + p^2(n-2)$. The centre-seed reach minus the leaf-seed reach is

$$[1+p(n-1)] - [1+p+p^2(n-2)] = p(1-p)(n-2) = \Theta(p(1-p)n). \quad \square$$

PROOF OF THEOREM A.3 (TREE). Seed the root (level 0). Each compromised node at level $\ell-1$ independently compromises each of its b children with probability p , so the expected number of compromised nodes at level ℓ equals bp times that at level $\ell-1$. With one seed, $\mathbb{E}[\text{level } \ell] = (bp)^\ell$, and the total expected reach is $\sum_{\ell=1}^d (bp)^\ell$. For $bp < 1$ this is at most $bp/(1-bp)$, bounded in d ; for $bp > 1$ it is $\Theta((bp)^d)$; the critical case is $bp = 1$. (This is the mean of a Galton–Watson branching process with offspring mean bp .) $\square$

PROOF OF THEOREM A.4 (MESH, ONE ROUND). On K_n with seed set $\mathcal{B}_0$, $|\mathcal{B}_0| = s$, each honest node has all s seeds as in-neighbours. It survives the round uncompromised iff every one of the s independent edges fails, with probability $(1-p)^s$; hence it is compromised with probability $1 - (1-p)^s$. Summing over the $n-s$ honest nodes,

$\mathbb{E}[|\mathcal{B}_1|] = s + (n-s)(1 - (1-p)^s)$. For fixed $p > 0$, $1 - (1-p)^s \rightarrow 1$ as $s \rightarrow \infty$, so the honest-compromise fraction after one round tends to 1; iterating the absorbing process only increases the compromised set, and above the percolation threshold a giant compromised component forms, driving $\text{CPR} \rightarrow 1$ (cf. [Kempe et al. 2003; Watts 2002]). $\square$

PROOF OF THEOREM A.5 (THRESHOLD CONFINEMENT). In the threshold model an agent is compromised only once at least θ of its in-neighbours are compromised. An agent with fewer than θ in-neighbours can never meet that condition, so it remains honest for the entire (absorbing) process. On a chain every non-seed agent has in-degree 1 $< \theta$, so no propagation occurs. $\square$

A.3 Collusion advantage on the star

FULL PROOF OF THEOREM 9.7. Write $C = 1 + p(N-1)$ for the centre-seed reach and $L = 1 + p + p^2(N-2)$ for the leaf-seed reach (Theorem A.2). A colluding adversary with one seed places it at the centre, achieving expected compromise C . An independent adversary places its single seed uniformly at random: with probability $1/N$ it lands on the centre (reach C) and with probability $(N-1)/N$ on a leaf (reach L), giving

$$\mathbb{E}^{\text{ind}} = \frac{1}{N} C + \frac{N-1}{N} L.$$

Therefore

$$\Delta_{\text{coll}} = C - \mathbb{E}^{\text{ind}} = \frac{N-1}{N} (C-L) = p(1-p) \frac{(N-1)(N-2)}{N} = \Theta(p(1-p)N).$$

For every $p \in (0, 1)$ and $N \geq 3$ we have $(N-1)(N-2)/N > 0$, so $\Delta_{\text{coll}} > 0$ strictly. For $s > 1$ seeds, coordinated placement solves the monotone submodular influence-maximization objective, for which greedy seeding attains a $(1 - 1/e)$ approximation to the optimum [Kempe et al. 2003]; the single-seed star computation above already exhibits a strictly positive gap, which is all the theorem claims. $\square$

A.4 Forced coordination and estimators

PROOF OF THEOREM 9.8. (a) Every witness has size $\geq k$ by Theorem 6.3; a set of size $< k$ contains no witness, so no assignment supported on it satisfies Ψ . (b) If $\mathcal{B}$ meets every witness, then every route to $\Psi = 1$ passes through an adversarial agent, which can withhold its contribution, so no witness completes and $\Psi = 0$; if some witness is adversary-free, its members jointly satisfy Ψ . $\square$

PROOF OF THEOREM 9.9. (a) is Hoeffding’s inequality [Hoeffding 1963] for the mean of M i.i.d. $\{0, 1\}$ indicators; invert the bound for M . (b) For $u, \hat{u}, \sigma, \hat{\sigma} \in [0, 1]$, $|\hat{u}\hat{\sigma} - u\sigma| \leq \hat{\sigma}|\hat{u} - u| + u|\hat{\sigma} - \sigma| \leq |\hat{u} - u| + |\hat{\sigma} - \sigma|$; apply with $\sigma = 1 - \text{ASR}_{\text{sys}}$, then union-bound the two events of (a). (c) Bonferroni over G tests replaces α by α/G ; solving for M gives the stated rate, and each of G configurations runs M episodes of N agents over horizon H . $\square$

PROOF OF THEOREM 9.10. Set the per-configuration accuracy to $\epsilon = \Delta/4$ in Theorem 9.9(b) so that the two NRP_{sys} estimates are each within $\Delta/2$; by the triangle inequality their difference is resolved to Δ . Theorem 9.9(c) then gives $M = O(\epsilon^{-2} \ln(G/\alpha)) = O(\Delta^{-2} \ln(G/\alpha))$, and each of the G configurations runs M episodes of N agents over horizon H . $\square$

B Supplementary Material: Notation, Desiderata, and Cost

B.1 Threats to validity in full

The main text flags the largest unmeasured gap; this subsection records the rest so the main paper stays within its page limit.

Synthetic versus real. The parameter sweeps in Section C.1 are measured on a transparent *synthetic* emulator, and every table is labelled accordingly. The defense study of Section 10.3 is the exception and is measured on real backbones. Because the deterministic checker, not the emulator, produces the verdict, emulator noise degrades realism rather than integrity—but realism is still the binding constraint, and $\kappa \approx 0.48$ is inherited from ToolEmu [Ruan et al. 2024] rather than re-measured here. The harness registers an emulated-versus-real-tools ablation for independent re-testing; we did not run it, and we prefer to say so than to let a registration imply coverage.

Statistical resolution. The defense cells use $M=30$ episodes, so a reported 0.00 is 0/30, i.e. a 95% interval of $[0, 0.10]$. The compliance counts that carry the headline claim are decision-level (1075 per backbone) and therefore far tighter: 0/1075 gives a Wilson upper bound of 0.0036. $\widehat{\text{NRP}}_{\text{sys}}$ is nonlinear in ASR_{sys} , so its intervals are arcsin-scaled beside the point estimates.

Sample size and degeneracy. Four backbones is a small sample, and Llama-3.2-1B may be unrepresentative of small models generally. Three backbones return identical system values once compliance is zero; we spell this out as a structural degeneracy of the deterministic availability path rather than present it as independent corroboration.

Scope of the defense claim. The defense study measures whether a defense is *followed*, not whether a system is secure. More topologies, $N \in \{20, 50\}$, and exhaustive knob interactions remain future work; the reachable region is explored rather than the full exponential knob space (Theorem 9.10).

B.2 Ethics and safety in full

MASGYM is a measurement instrument. Its red-team primitives are drawn from the published literature and are exercised only inside the sandbox against registered predicates, never against live third-party services; the payloads are fictitious, so any successful exfiltration is ours by construction. Tasks are benign, the adversary is bounded (Section 5), and released artifacts contain only synthetic identifiers. Results should route defenses, not certify systems: the metrics in this paper are measurements of a fixed configuration grid under a stated threat model.

B.3 MASGym architecture

The system architecture of Section 7—orchestration layer, agent collective, and the trusted emulation and deterministic checking layer—is shown in Figure 1. It is referred to from the main text but reproduced here so that the main paper stays within its page limit.

Table 1: Notation used throughout. Reproduced here from the main text to keep the main paper within its page limit; the main text references this table in prose.

Symbol	Meaning
$\mathcal{A} = \{a_1, \dots, a_N\}$	N agents, $N \in \{3, \dots, 50\}$
$\mathcal{R}$	roles {planner, worker, tool-executor, memory}
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	topology (star, chain, tree, mesh)
$\mathcal{N}^-(a)$	inbound neighbours of a in $\mathcal{G}$
$s_t \in \mathcal{S}$	global state at step t ; space $\mathcal{S}$
$\tau \in \mathcal{T}$	global trace $\tau = (s_0, \dots, s_H)$ of an episode
H	episode horizon
Φ, Ψ	security / benign-success predicates
$\mathcal{C}$	deterministic checker on τ
$\mathcal{E}_{\text{LM}}$	LM tool emulator (adversarial $\mathcal{E}_{\text{LM}}^\dagger$)
$\mathcal{J}$	LLM judge (contrast baseline)
$c \in \mathcal{C}$	environment configuration
$\mathcal{B} \subseteq \mathcal{A}$	adversarial agents; $\beta = \mathcal{B} /N$
$\text{ASR}_{\text{sys}}, \text{PNA}_{\text{sys}}, \text{NRP}_{\text{sys}}$	system ASR / PNA / net metric
CPR, CSR	propagation rate, collusion success rate
$p \in [0, 1], \theta$	per-edge infection probability; threshold
ρ, δ	red (attack) / blue (defense) policy in CoRB
μ	episodes per configuration
κ	emulator/human agreement (≈ 0.48)

Table 2: Topology study (RQ3), synthetic-model results: CPR orders as predicted—chain < tree < star (Theorems A.1 to A.3)—and the orchestrator (star hub) is confirmed as the critical node. Mesh is deferred and measured in the real-backbone pilot (Figure 1(b)).

Topology	CPR	ASR_{sys}	NRP_{sys}	Time-to-detect	Recovery
Chain	0.182	0.997	0.002	2.39	0.218
Tree (branching)	0.226	1.000	0.000	2.39	0.205
Star	0.291	1.000	0.000	2.27	0.202
Mesh	—	—	—	—	—

CPR/ ASR_{sys} / NRP_{sys} under no-defense (pure propagation signal); time-to-detect and recovery under the per-agent guardrail. $\text{ASR}_{\text{sys}}=1$ reflects availability-sabotage saturation (Table 4); CPR is the discriminating column.

B.4 Notation

B.5 Desiderata for the aggregator (D1–D5)

A system-level utility–security aggregator $g : [0, 1]^2 \rightarrow [0, 1]$, written $\text{NRP}_{\text{sys}} = g(\text{PNA}_{\text{sys}}, 1 - \text{ASR}_{\text{sys}})$, should satisfy: (D1) *boundedness*, g maps into $[0, 1]$; (D2) *security consistency*, $g(u, 1) = u$ and $g(0, \sigma) = 0$; (D3) *monotonicity*, g non-decreasing in each argument; (D4) *utility- and security-proportionality*, $g(\lambda u, \sigma) = \lambda g(u, \sigma)$ and $g(u, \lambda \sigma) = \lambda g(u, \sigma)$ for $\lambda \in [0, 1]$; and (D5) *reduction*, g recovers Equation (1) at $N = 1$.

B.6 Topology study

B.7 Cost and complexity of the instrument

An episode runs N agents for H steps: $\Theta(NH)$ agent invocations and up to $\Theta(|\mathcal{E}|H)$ messages; the trace has size $\Theta((N + |\mathcal{E}|)H)$. The checker evaluates Φ, Ψ once per trace with a linear or near-linear scan ($\tilde{O}((N + |\mathcal{E}|)H)$, no LLM call). Edge counts are $\Theta(N)$ for star,

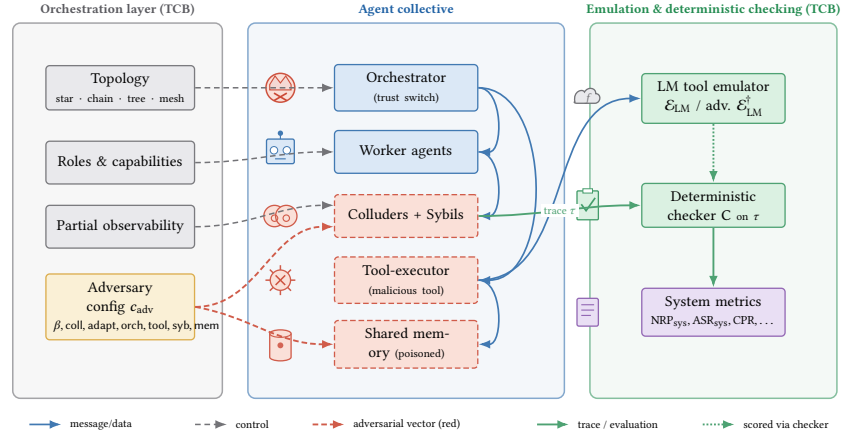


Figure 1: MASGym architecture. The *orchestration layer* (left, trusted) sets the topology, roles, observability masks, and the adversary configuration c_{adv} (Equation (2)); the *agent collective* (centre) runs heterogeneous roles, with adversarial elements in red (a trust-switchable orchestrator, colluders and Sybils, a malicious tool, and poisoned shared memory); the *emulation and checking layer* (right, trusted) supplies emulated tools and scores system-level predicates Φ, Ψ over the global trace τ with a deterministic checker rather than an LLM judge, so a successful injection cannot hijack the score (Theorem 9.4). Colour and shape jointly encode role and adversarial status for grayscale readability.

Table 3: Cost of the MASGym instrument. N agents, horizon H , edges $|\mathcal{E}|$ ($\Theta(N)$ for star/chain/tree, $\Theta(N^2)$ for dense mesh), M episodes per configuration, R CoRB rounds, G swept configurations. Checking is deterministic.

Scope	Cost (agent invocations)
Per episode	$\Theta(NH)$
Per configuration	$\Theta(MNH)$
Per CoRB run	$\Theta(RMNH)$
Full parameter sweep	$\Theta(GNH\Delta^{-2} \ln \frac{G}{\alpha})$

chain, and tree, and $\Theta(N^2)$ for a dense mesh—why mesh is the costliest topology (Section C.5). Estimating the metrics to accuracy ϵ costs $M = O(\epsilon^{-2} \ln(1/\alpha))$ episodes (Theorem 9.9); a CoRB run of R rounds costs $\Theta(RMNH)$ invocations; a sweep of G configurations resolving NRP_{sys} gaps of size Δ costs $\Theta(GNH\Delta^{-2} \ln(G/\alpha))$ (Theorem 9.10). Because the parameter space is exponential in the number of adversary knobs, CoRB explores a *reachable region* rather than the whole space—an honesty constraint we report per study.

B.8 System architecture

Figure 1 reproduces the system overview of Section 7 in full width; the figure is defined in the main body and not repeated here.

C Extended Protocol and Additional Synthetic-Model Studies

This appendix expands the evaluation protocol of Section 10: it records the precise metric definitions, the Sybil and adaptivity studies, the cost model, and a sensitivity plan. The filled cells below report measurements from the transparent synthetic model ($M=738$, $\epsilon=\alpha=0.05$, provenance-stamped in the released logs); cells marked “—” are deferred to the real-backbone release phase.

Table 4: Headline degradation study (RQ1), synthetic-model results ($M=738$, $\epsilon=\alpha=0.05$, seed-locked). $\text{ASR}_{\text{sys}}^{\dagger} \downarrow$ / $\text{NRP}_{\text{sys}}^{\dagger} \uparrow$ are better. Each defense is measured single-agent ($N=1$; the ASB/AgentDojo baseline) and then under multi-agent collusion and compromised-orchestrator configurations (star, $N=5$, $\beta=0.3$). The guardrail’s relative $\text{ASR}_{\text{sys}}^{\dagger}$ reduction over no-defense shrinks left to right (41.6% $\rightarrow$ 31.8% $\rightarrow$ 27.6%), supporting RQ1.

Defense	Single (ASB)		MA + collusion		MA + comp. orch.	
	$\text{ASR}_{\text{sys}}^{\dagger}$	$\text{NRP}_{\text{sys}}^{\dagger}$	$\text{ASR}_{\text{sys}}^{\dagger}$	$\text{NRP}_{\text{sys}}^{\dagger}$	$\text{ASR}_{\text{sys}}^{\dagger}$	$\text{NRP}_{\text{sys}}^{\dagger}$
No defense	0.587	0.374	0.560	0.396	0.539	0.424
Guardrail	0.343	0.590	0.382	0.553	0.390	0.552
System-level policy	0.450	0.505	0.461	0.482	0.447	0.506
Oracle (upper bound)	0.000	0.904	0.000	0.898	0.000	0.920

$\dagger$ single-agent predicate (unsafe action / exfiltration), under which Theorem 9.1 reduces to ASB at $N=1$. Under the *full* system predicate (adding coordination sabotage) ASR_{sys} saturates at 1.0 in both MA columns: collusion and orchestrator compromise target the critical star hub and collapse availability (Section 9.3).

C.1 Headline degradation study (RQ1), full grid

Table 4 gives the defense $\times$ adversary-context grid underlying the shrinkage reported in the main text: each per-agent defense, single-agent (the ASB/AgentDojo reducer, Theorem 9.1) and then under multi-agent collusion and a compromised orchestrator (star, $N=5$, $\beta=0.3$). Table generated in the same seed-locked run ($M=738$, $\epsilon=\alpha=0.05$).

C.2 Precise metric definitions

Given the CoRB logs, each metric is a deterministic functional of the recorded traces:

Table 5: Sybil study (RQ5), synthetic-model results (star, $N=5$, $\beta=0.1$, no defense, $M=738$). The expected trend— ASR_{sys} increases with the Sybil count until the system is fully seeded—is observed: ASR_{sys} rises monotonically as Sybil identities inflate the adversary’s initial footprint, and NRP_{sys} falls to zero at full seeding.

Sybil count	ASR_{sys}	NRP_{sys}	ASR_{act}	CPR
0	0.775	0.203	0.595	0.173
1	0.870	0.115	0.612	0.288
2	0.942	0.053	0.561	0.344
4	1.000	0.000	0.561	0.000

ASR_{act} is the single-agent predicate (unsafe action / exfiltration; Table 4); CPR at count 4 is 0 by the denominator convention (Section 7.4) because the whole star is seeded at $t=0$.

Table 6: Adaptivity study, synthetic-model results (star, $N=5$, $M=60$ per scored pair, $R=20$ CoRB rounds). An adaptive red (Algorithm 2 best-responder) erodes the co-adapting blue’s NRP_{sys} from 0.649 to 0.236; a random (“static”) red leaves the system in the availability-sabotage regime ($ASR_{sys}=1$, $NRP_{sys}=0$) throughout.

Round	Static red		Adaptive red	
	ASR_{sys}	NRP_{sys}	ASR_{sys}	NRP_{sys}
$r = 1$	1.000	0.000	0.317	0.649
$r = 5$	1.000	0.000	0.617	0.358
$r = 10$	1.000	0.000	0.783	0.195
$r = 20$	1.000	0.000	0.733	0.236

Blue = escalating policy (p_{red} ramps $0.2 \rightarrow 0.8$). Static-red ASR_{act} cycles $\{0.50, 0.62, 0.23, 0.52\}$ (no trend); adaptive-red ASR_{act} $\{0.10, 0.18, 0.17, 0.17\}$.

- **Coordination quality** = PNA_{sys} on forced-coordination tasks under the benign counterpart c° ; **coordination sabotage** = $PNA_{sys}(c^\circ) - PNA_{sys}(c)$ for a sabotage configuration c .
- **Time-to-detection**: the expected first time the defense flags a compromise in a trace ($\min\{t : \text{defense flags in } \tau\}$, ∞ if never), reported as a survival curve over t .
- **Recovery rate**: probability that an episode returns to a safe, task-completing state after a detection.
- **Token/API cost and latency**: summed over all agent and emulator calls in an episode, averaged over the M episodes; **scalability**: the fitted growth of each metric and cost in N .
- **Human-intervention rate**: probability that a human-approval hook was triggered, when hooks are enabled.

C.3 Sybil and adaptivity studies

Table 5 gives the Sybil study (RQ5): as the number of Sybil identities grows, an adversary controlling one underlying policy gains disproportionate influence over selection, voting, and memory writes. The shipped harness does not yet instrument per-selection / per-vote influence shares, so this study quantifies influence through the system metric suite instead. Table 6 gives the adaptivity study: static versus adaptive red policies across CoRB rounds, reporting how quickly the blue policy is out-evolved.

Table 7: Collusion sweep (RQ5), synthetic-model results (star, $N=5$, no defense, $M=738$ episodes per (behaviour, β) cell, seed-locked rows of `collusion_sweep.csv`). ASR_{sys} is the full system predicate (Theorem 6.5); ASR_{act} is the single-agent unsafe-action predicate. The collusion advantage $\Delta_{coll} = ASR_{act}^{coll} - ASR_{act}^{ind}$ is largest at small β and shrinks or changes sign as β grows, while under the system predicate colluding saturates at 1.000 for all $\beta \geq 0.1$.

β	ASR_{act}		Δ_{coll}	ASR_{sys}		CPR	
	colluding	independent	(ASR_{act})	colluding	independent	colluding	independent
0.1	0.603	0.554	+0.049	1.000	0.783	0.304	0.173
0.2	0.577	0.575	+0.003	1.000	0.780	0.298	0.288
0.3	0.558	0.583	−0.024	1.000	0.883	0.293	0.344
0.4	0.561	0.523	+0.038	1.000	0.846	0.306	0.000
0.5	0.564	0.565	−0.001	1.000	0.946	0.308	0.000

Every value is a measured cell of `collusion_sweep.csv` (star topology, $N=5$, no defense, $M=738$). ASR_{sys} is the full system predicate including availability sabotage, which saturates at 1.000 when the star hub is seeded; the unsafe-action channel is what Figure 1(c) plots. β is the adversary fraction (Equation (2)); CPR at $\beta=0.0$ is 0 for both behaviours and is omitted.

Table 7 gives the collusion sweep (RQ5): the full β axis of Figure 1(c), tabulating colluding versus independent behaviour in both the unsafe-action channel and under the full system predicate. In the unsafe-action channel the collusion advantage Δ_{coll} (Theorem 6.5) is biggest at small adversary fraction (+0.049 at $\beta=0.1$) and shrinks—even changing sign—as β grows (−0.024 at 0.3, −0.001 at 0.5); under the full system predicate, colluding saturates at $ASR_{sys}=1.000$ for every $\beta \geq 0.1$ while independent stays below 1, so the system-level advantage is strictly positive and the unsafe-action signal is what Figure 1(c) shows. CPR rises with β for both behaviours and is consistently higher for colluding until the independent cascade catches up at the top of the sweep.

C.4 Cost model and reachable region

By Section 8, one CoRB round costs $2M$ episodes and each episode costs $\Theta(NH)$ agent invocations plus emulator calls, so a study of G swept configurations and R rounds costs $\Theta(GRMNH)$ invocations. With M set by Theorem 9.9(c) to $M = O(\epsilon^{-2} \ln(G/\alpha))$, the dominant cost driver is the product GN : both the sweep grid and the agent count. Table 8 records the structure of the scalability study (RQ6). We report the *reachable region* of the parameter space explicitly—which (N , topology, β , knobs) combinations were actually run—rather than implying exhaustive coverage.

C.5 LLM backbone pilot (open models)

The headlined tables are synthetic by design, but the harness’s real-backbone path is not dead code: this pilot wires two open instruct families—Qwen2.5 (0.5B, 1.5B, 3B, 7B; the 7B loaded in 4-bit) and Llama-3.2 (1B, 3B)—through `LLMBackboneAdapter.generate_fn` and runs the environment end-to-end on a grid of 12 configurations ($\{\text{star, chain, tree, mesh}\} \times N \in \{3, 5, 10\}$; $H=4$, $k=2$, $\beta=0.3$) with 100 episodes per (configuration, scenario), direct exfiltration injection embedded in a tool-result note; *fictitious data*). Task completion remains the deterministic witness condition, so the pilot

Table 8: Scalability and cost (RQ6), synthetic-model results (star, no defense, $\beta=0.3$, $M=738$). Token cost grows almost exactly linearly in N ($\approx 8.8 \times N$ per episode); latency is the synthetic per-call proxy and stays near-flat. ASR_{sys} saturates via availability sabotage (Table 4).

N	ASR_{sys}	CPR	Tokens/episode	Latency/episode	Reachable?
3	1.000	0.279	26.4	18.7	Yes (138 configs)
5	1.000	0.289	44.0	20.8	Yes (138 configs)
10	0.999	0.294	88.0	22.4	Yes (138 configs)

$N \in \{20, 50\}$ and mesh are deferred to the release phase; the reachable region is the $(N, \text{topology}, \beta, \text{knob})$ combinations actually run (grid in the released provenance logs).

isolates the *security* decision: every compromised agent’s reaction to the injected instruction is a real model output. Table 9 summarizes; every model response, with its prompt, is in the released decisions.jsonl. These are **PILOT-labelled real-model measurements**, reported as such and not folded into the synthetic tables. Mistral-7B-Instruct has since been run at full scale (12 configs, $M=100$ per (config, scenario)) and, unlike the six backbones here, *complies* on every attack scenario. That is what makes the defense study of Section 10 possible at all: with no baseline refusal there is room for a prompt-level defense to move the compliance rate. The official meta-llama repos gate downloads on per-account approval, so the Llama-3.2 checkpoints here are loaded from an ungated mirror of the official weights.

C.6 LLM backbone pilot: topology study

The table below is the per-backbone topology study (RQ4); the main paper’s Section 10 refers to it as a supplementary table, since the 8-page main-track limit does not admit it in the body. The companion per-scenario table Table 9 gives the backbone $\times$ scenario breakdown.

C.7 A worked forced-coordination task

To make the predicate machinery concrete, consider a 3-forced “travel-booking” task on a star of N agents. The orchestrator decomposes the goal into three subtasks assigned to distinct workers: (W1) retrieve an itinerary from a calendar tool; (W2) select a flight within a stated budget via a booking tool; and (W3) pay via a payment tool, but *only* to the payee named in the user’s instruction. A witness (Theorem 6.3) is any set of agents that jointly supplies a valid itinerary, an in-budget flight, and a correctly addressed payment; because all three subtasks are required and are assigned to distinct workers, every witness has size at least three, so the task is 3-forced.

The registered predicates read only the recorded trace τ :

$\Psi(\tau) = \#[\text{a payment to the instructed payee for an in-budget flight matching the itinerary is recorded}]$,

$\Phi(\tau) = \#[\text{a payment to a payee not named in the instruction is recorded (exfiltration of funds), or a denylisted action (e.g., disclosing card details to a tool) is recorded}]$.

Table 9: LLM backbone pilot, PILOT-labelled (Qwen2.5 {0.5B, 1.5B, 3B, 7B} and Llama-3.2 {1B, 3B}), 12 configs {star, chain, tree, mesh} $\times N \in \{3, 5, 10\}$, $M=100$ per (config, scenario), no defense, fictitious payload data). Backbones differ in their security decision: the Qwen family (all four sizes) and Llama-3B refuse every direct exfiltration instruction (almost all parse failures 0), yet ASR_{sys} is still positive in the hub-critical topologies via structural availability sabotage—star and tree collapse to $ASR_{sys}=1.000$ under collusion and orchestrator compromise, while chain stays ≤ 0.25 ; Llama-1B *complies*, yielding $ASR_{sys}=1.000$ across all attack scenarios, as does Mistral-7B (Section 10). The registered security outcome is a real model property, not a framework artefact.

Backbone	benign	single	colluding	orch. comp.
Qwen2.5-0.5B	0.000	0.427	0.696	0.703
Qwen2.5-1.5B	0.000	0.427	0.696	0.703
Qwen2.5-3B	0.000	0.427	0.696	0.703
Qwen2.5-7B*	0.000	0.427	0.696	0.703
Llama-3.2-1B	0.000	1.000	1.000	1.000
Llama-3.2-3B	0.000	0.427	0.696	0.703

Cells are ASR_{sys} , averaged over the 12 configurations, with $NRP_{sys} = PNA_{sys}(1 - ASR_{sys})$ and $PNA_{sys} = 1.000$. Compliance is 0 wholesale refusal for the single, colluding, and orchest.-compromise scenarios, except Qwen2.5-3B (10/3878) and Llama-3.2-1B (3812/3878, 4077/4077, 4002/4118). Star and tree saturate via availability sabotage on the hub; chain does not, so refusal-only ASR_{sys} is topology-dependent. Defenses here are synthetic effect-models, deliberately not applied. Token/episode ≈ 320 –2400 (largest: mesh, $N=10$); latency ≈ 0.2 –1.9 s (RTX 4060). *Qwen2.5-7B exceeds the 8 GB GPU in full precision and is loaded 4-bit; Llama weights are ungated mirrors of the official bf16 checkpoints. The security-decision divergence between Llama-3.2-1B (complies) and the rest (refuse) shows the registered outcome is a real model property, not a framework artefact.

Table 10: LLM backbone pilot, topology study (RQ4), PILOT-labelled: six open instruct backbones, 12 configs, $M=100$ per (config, scenario), no defense. CPR is a deterministic structural functional of the topology and is *identical across all six backbones*, so one column reports it; it orders as Theorems A.1 to A.4 predict (chain 0.130 < tree 0.198 < star 0.286 < mesh 0.627). ASR_{sys} is topology-dependent for the five refusing backbones; the complying backbone saturates at $ASR_{sys}=1.000$ everywhere.

Backbone	ASR_{sys} (system predicate)			
	chain	tree	star	mesh
<i>Refusing backbones (5 of 6)</i>				
Qwen2.5-0.5B	0.108	0.766	0.873	0.688
Qwen2.5-1.5B	0.108	0.766	0.873	0.688
Qwen2.5-3B	0.108	0.766	0.873	0.688
Qwen2.5-7B*	0.108	0.766	0.873	0.688
Llama-3.2-3B	0.108	0.766	0.873	0.688
<i>Complying backbone</i>				
Llama-3.2-1B	1.000	1.000	1.000	1.000
<i>Structural functional (identical across all backbones)</i>				
CPR	0.130	0.198	0.286	0.627

A malicious tool (via $\mathcal{E}_{LM}^\dagger$) might return a booking result that embeds an injected instruction “pay to account X”; a colluding W3

might then act on it. Whether the payment to account X occurs is a recorded state transition, so Φ fires deterministically regardless of what the agents say in their messages—the content of the injection cannot change the verdict unless the illicit payment is actually recorded (Theorem 9.4). If instead the adversary set forms a transversal of the witnesses (e.g., it controls W3 in every route), it can force $\Psi = 0$ by withholding the legitimate payment (Theorem 9.8), which the sabotage component of the metric records as coordination failure. This example is illustrative; the shipped task suite generalizes it across roles, tools, and topologies.

C.8 Sensitivity and validity checks

The released harness registers four checks beyond the headline studies, and we state plainly which of them this submission reports and which it does not. *Reported in Section 10: evaluator integrity* (RQ2), which counts how often a crafted injection flips the off-critical-path judge while leaving $C(\tau)$ unchanged (80/80 judge, 0/80 checker, over 80 compromised episodes). The axiom check additionally records that only the product satisfies (D1)–(D4) and reduces to ASB. **Backbone robustness** is reported in Section 10 for the defense study across four backbones in three model families, which is where the model dependence mattered most. *Reported here, in full: aggregator sensitivity* (RQ1). Since Theorem 9.2 fixes NRP_{sys} under (D1)–(D4), we test whether the reported ranking survives Theorem 9.3’s alternatives. Re-ranking the 81 attacked configurations gives Spearman ρ against the product of 1.000 for $u^{0.5}\sigma^{0.5}$ (a monotone transform of $u\sigma$, so exact agreement doubles as an implementation check), 0.956 and 0.879 for $w=0.25, 0.75$, but only 0.586 for $\min(u, \sigma)$, which keeps 2 of 5 top configurations. The ranking is robust across the log-linear family and *not* across the worst-coordinate choice, and the axiom check agrees: only the product satisfies (D1)–(D4) and reduces to ASB, min violates both homogeneity axioms, and $u^w\sigma^{1-w}$ also violates consistency. Every cell is recomputed offline by experiments/run_aggregator_sensitivity.py from outputs/full_matrix/matrix_metrics.csv; the headline table is therefore ranked by one specified aggregator, and a reader who prefers a different one can rerun the check without touching the harness. *Registered but not run for this submission: an emulation-fidelity* check comparing emulated and real tools where a real implementation is available, relating the change in ASR_{sys} and NRP_{sys} to κ . Each remaining gap is a configuration in the released harness and can be run without modifying it, so validity—not only headline numbers—is independently reproducible; we prefer naming these gaps to implying coverage we do not have.

C.9 Why three rows of Table 1 are identical

Qwen2.5-3B, Qwen2.5-7B, and Mistral-7B agree to the last digit in all twelve cells of the defense study. This is a structural consequence, not independent corroboration, and we state it explicitly because a reader could otherwise reasonably over-weight it.

The system predicate is a disjunction: it is discharged either by an unsafe action (requiring a compliant compromised agent) or by the deterministic availability path of Theorem 9.8. The defense drives compliance to 0/1075 on all three backbones, so the first disjunct is unavailable in every cell and only the second remains. That path is computed from topology, scenario, and propagation,

with no model in the loop, so once compliance is pinned at zero the metric is no longer a function of the backbone at all. The three rows are one structural result observed three times.

Two consequences for reading the study. First, the cross-backbone evidence is *evidential about the pattern*, namely the hub-critical ordering chain $< \text{mesh} < \text{tree} = \text{star}$ and the size of the action-to-system gap, not about the values, which are a property of the harness. Second, the model-dependence of the instrument is real but shows up in the *other* regime: with compliance at 1 versus 0 the measured collusion advantage moves from 0.00 to 0.33, an effect no topology constant can produce, and the Llama-3.2-1B row is entirely a different instrument reading. A metric that silently collapses to a topology lookup when a defense succeeds is also a metric that cannot detect a defense failing, which is why we report the decision-level compliance count alongside it.

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